



FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

BMIS2003 Blockchain Application Development

Assignment

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Tutorial Group: **Group 1**

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1.0 Introduction

Blockchain is a decentralised and transparent database system that allows for safe, secure transaction recording without central vendors (Nakamoto, 2025). The technology is designed to enhance data security, integrity, and efficiency in digital ecosystems, particularly where trust and transparency are crucial (Zhisheng, 2024). Smart contracts are self-executing programs kept on the blockchain that automatically enforce the terms of an agreement when certain criteria are satisfied (IBM, 2025). Together, blockchain and smart contracts eliminate the need for manual intervention, significantly improving transaction speed, accuracy, and reliability while minimising operational risk. These features enhance participant confidence and make the technology particularly suitable for applications requiring safe, transparent, and automated record-keeping (Mohd Javaid, 2025; Jell-Ojobor, Maria, 2025).

The integration of blockchain and smart contracts facilitates trustless transactions, reduces costs, and enhances transparency in real-time, making them well-suited for industries such as supply chain, finance, and real estate (Huan, N.T.Y, 2024). Real estate fractionalisation—breaking property ownership into smaller digital tokens—has emerged as a compelling use case. This approach allows multiple investors to co-own property through tokenization, reducing financial entry barriers and boosting asset liquidity (Lofty, 2025; RealIT, 2025). Research by Chintan (2022) emphasizes the effectiveness of blockchain in democratizing asset ownership and providing secure revenue-sharing mechanisms for digital investors.

Platforms like RealT and Lofty.ai utilize this concept to allow users to buy fractional shares in property and automatically earn rental income via smart contracts. Our project builds on this idea by developing a decentralized application (dApp) that allows users to view tokenized properties, invest with digital tokens, and automatically receive rental income. By incorporating blockchain and smart contracts, this system ensures security, transparency, and automation in property investment, aligned with the emerging needs of digital economies (Kuldeep, 2024). Our project intends to create a decentralised application (dApp) that will allow users to view tokenized properties, invest with digital tokens, and get rental revenue automatically through smart contracts, making real estate investment more accessible, secure, and efficient.

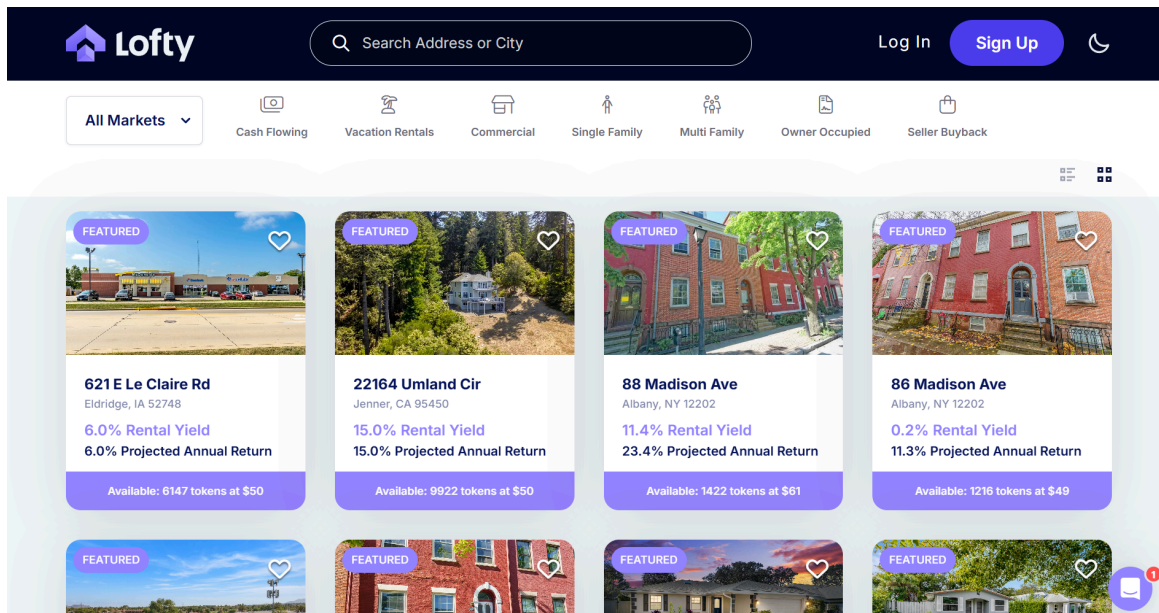


Figure 1.0.1 - Lofty.ai real estate platform

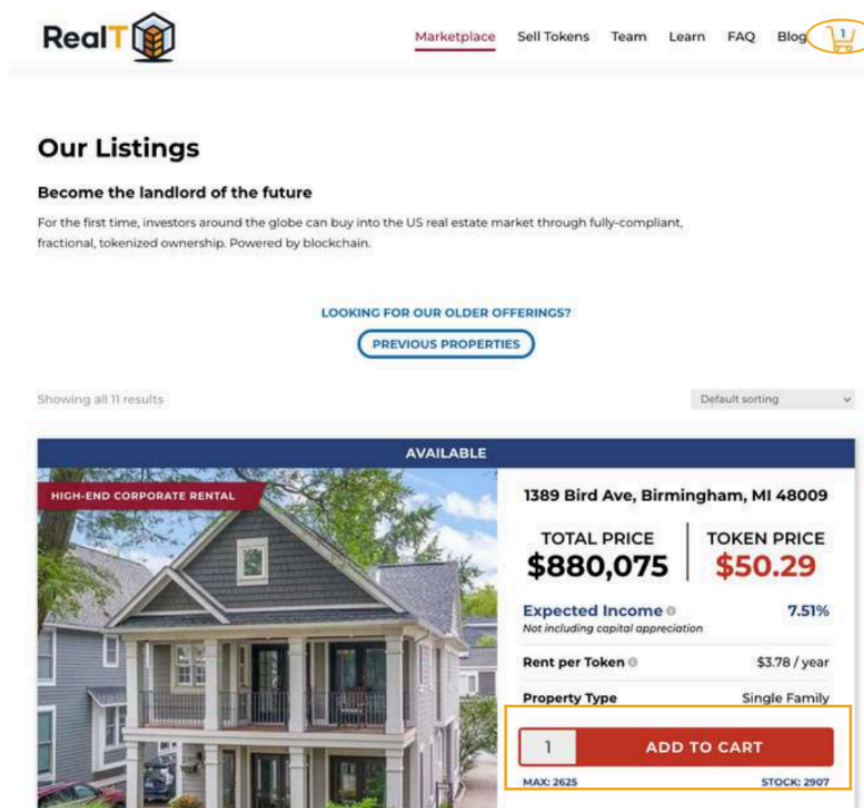


Figure 1.0.2 - RealT real estate platform

2.0 Problem Statement

1. Investing in traditional real estate usually requires a lot of capital and has low liquidity, making it difficult for most people to participate or quickly exit their investments.
2. Centralised real estate ownership platforms often have issues with transparency, trust, and limited automation in managing ownership and dividends. Unlike these centralised systems, decentralised applications (dApps) built on Ethereum smart contracts can tokenize real estate assets, allowing users to buy, sell, and trade fractional real estate shares with greater transparency, security, and efficiency.

3.0 Policy / Business Rules

Innova is a Blockchain-based real estate investment system which allows users to have a decentralised path to find the properties they want to invest on our website.

Policies

Transparency

All property listings, purchases, sell and token investments are recorded on the blockchain. This means that everything is forever recorded and visible to everyone. This promotes trust and prevents misunderstandings.

Security

Using smart contracts to ensure the safety of all transactions. These contracts operate automatically, protecting users from scams and mistakes.

Tokenization

Each property is divided into tokens, allowing people to acquire a little portion of it. These tokens can be tracked, traded, and used to establish ownership of assets on the blockchain.

Fair Access

Anyone can invest, but only administrators can add properties. Ensure that all users have equal possibilities to participate in investments while maintaining a controlled property upload process.

User Control

You have control of your own wallet and property tokens. No transaction can take place without your approval via your wallet.

Business Rules

Admin:

- Manage property listings for fractional ownership (add, update, delete).
- Set property details.
- Monitor transactions to ensure compliance.
- Verify and audit transactions and smart contract interactions.

Investor (User):

- Browse and filter investment properties.
- Register, authenticate, and connect wallets before investing.
- Invest by buying ownership tokens with blockchain records.
- Receive dividends based on property appreciation.
- View monthly rental income distribution for properties they've invested in.
- Stay updated on property changes and dividends.
- Buy and sell shares via the trading platform integrated with smart contracts.
- When a user sells shares:
 - The total shares owned by the user are reduced accordingly.
 - MetaMask may display the transaction as “-0.00 ETH” because it only shows ETH sent from the user, not ETH received to the user. This is a UI limitation, not a transaction issue.

4.0 Objectives

Objectives of this project create the decentralised application (dApp) platform for investor as follow:

- To use Ethereum smart contracts to create and deploy a decentralised application (dApp) for fractional ownership of real estate.
- To allow role-based access, secure user registration, and authentication (e.g., admin vs. investor).
- To enable administrators to list properties for fractional investment in real estate.
- To issue and administer tokens that stand for fractional ownership as well as real estate assets.
- To facilitate the transparent purchase, sale, and trading of fractional property shares by users.
- To incorporate a front-end interface that communicates with Ethereum testnet smart contracts that have been deployed.
- To automatically distribute dividends to users according to their percentage of shares.
- To store and show the history of user transactions within the dApp.

5.0 Contract Diagram

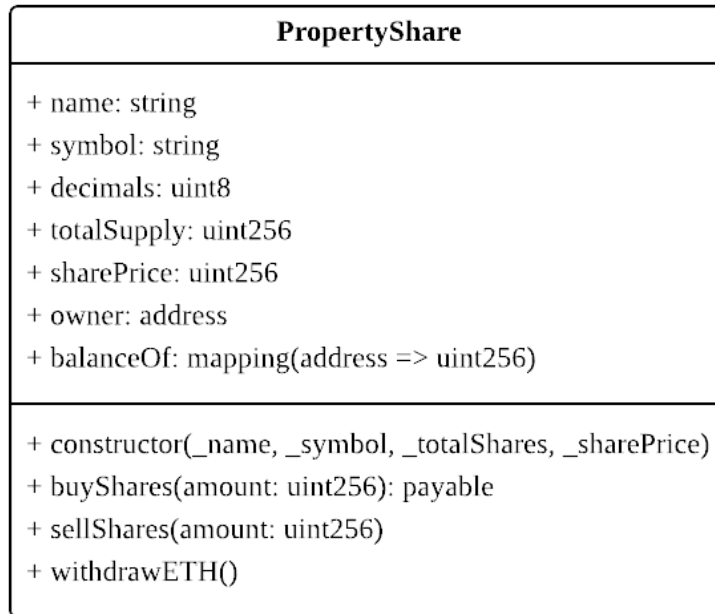
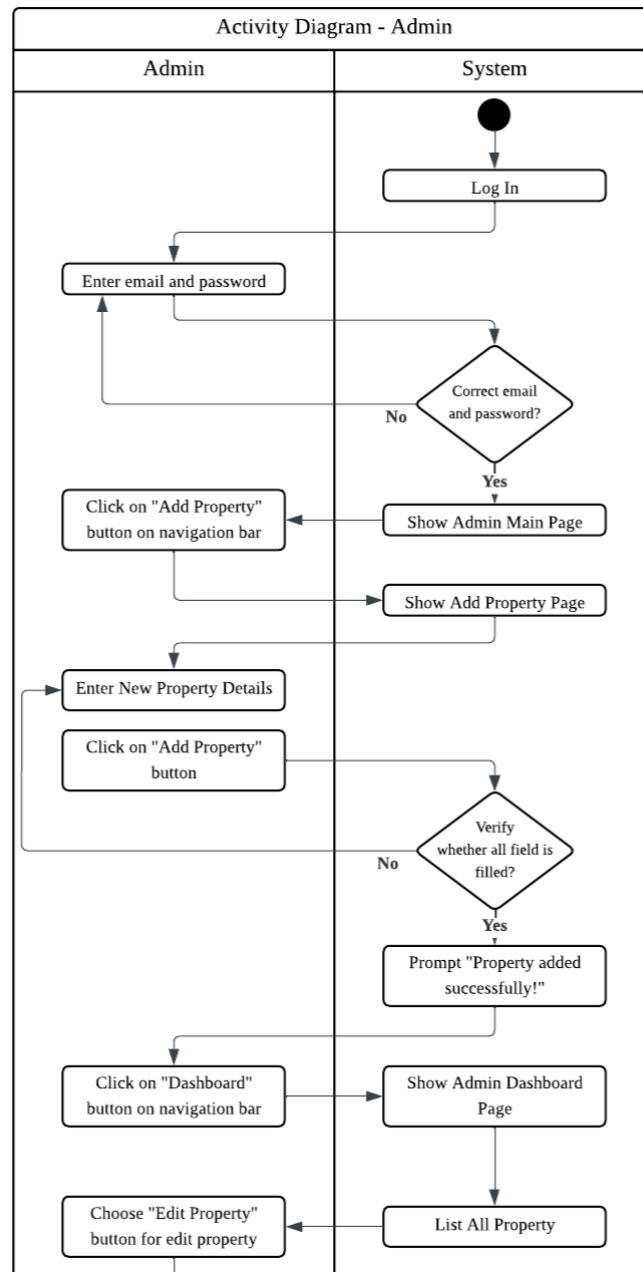


Figure 5.0.1 Contract Diagram - Innova

6.0 Activity Diagram

6.0.1 Activity Diagram - Admin



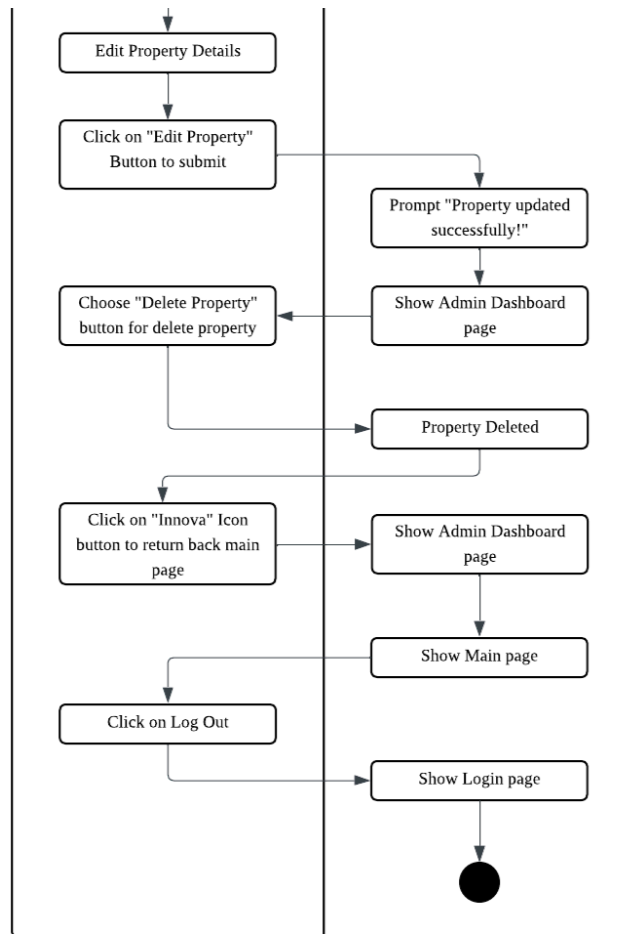
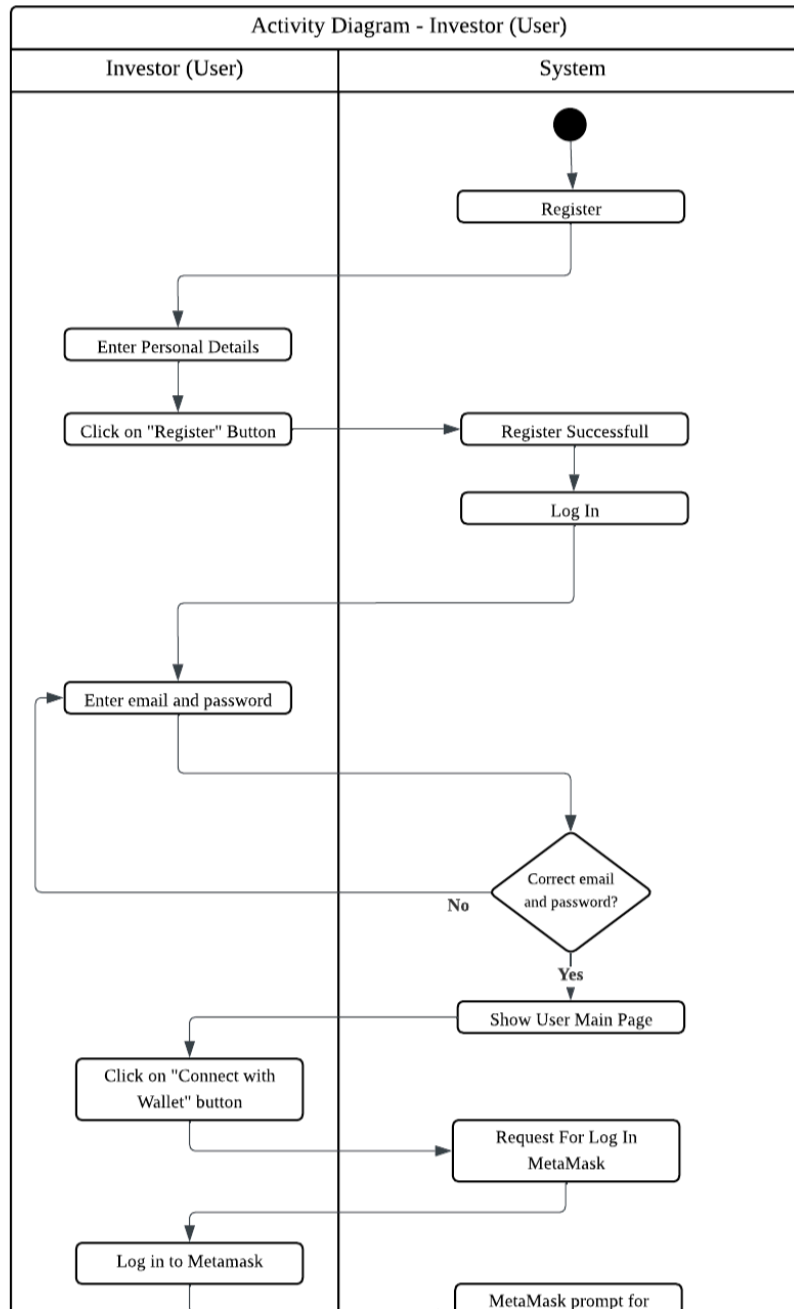
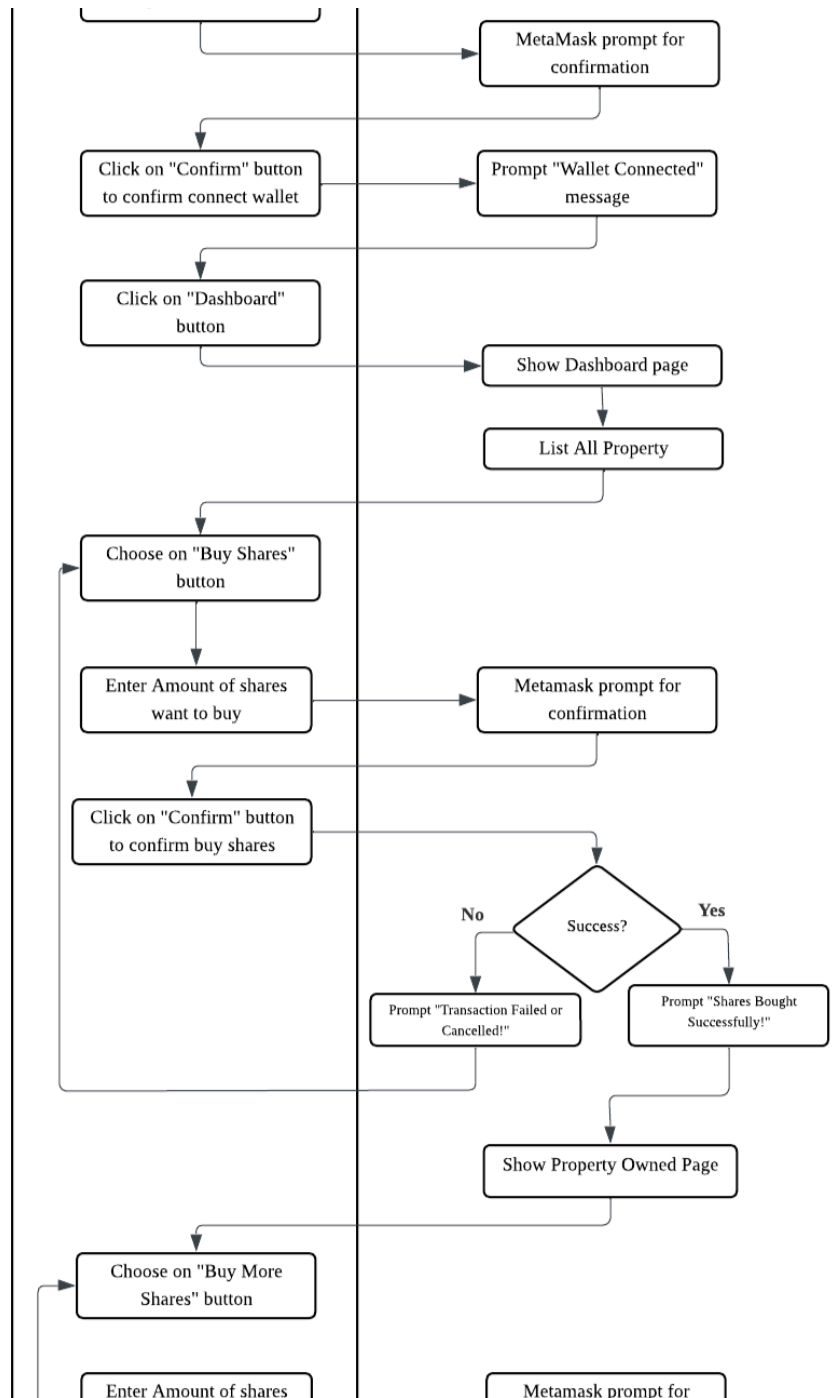
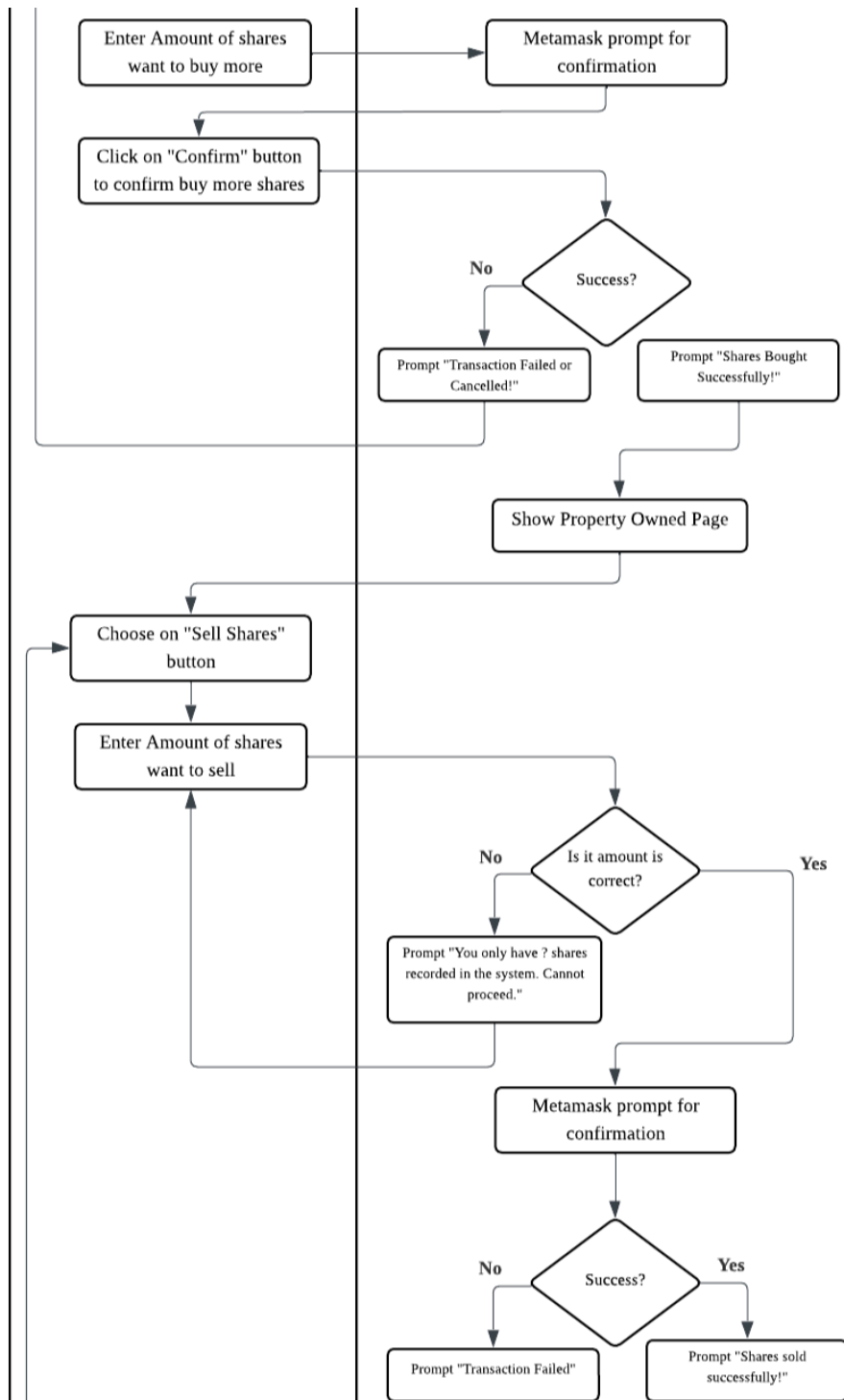


Figure 6.0.1.1 - Activity Diagram (Admin)

6.0.2 Activity Diagram - Investor (User)







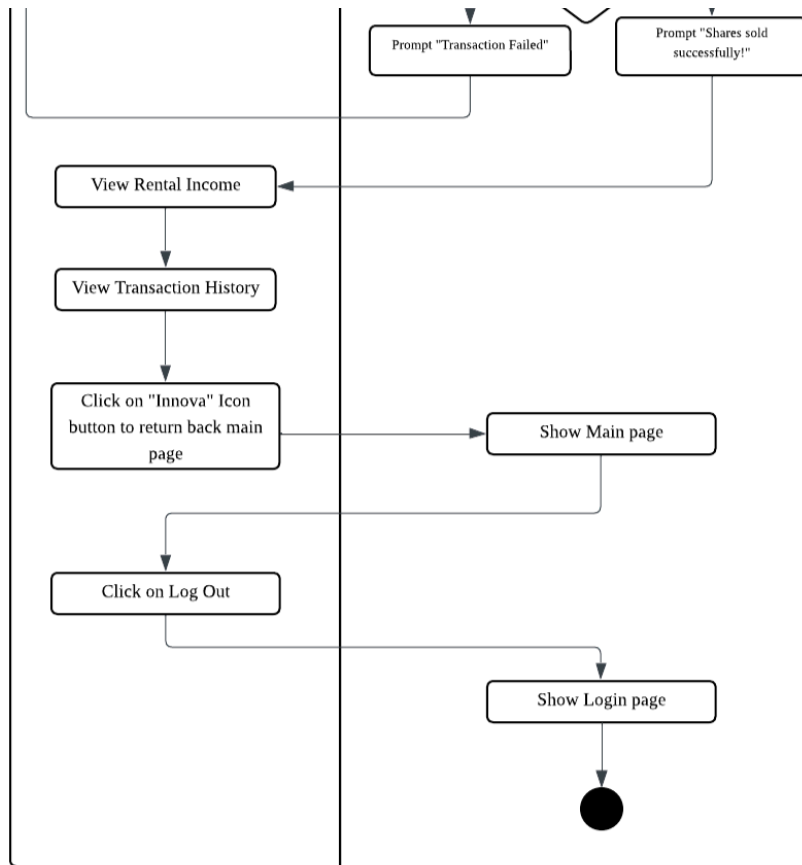


Figure 6.0.2.1 - Activity Diagram Investor (User)

7.0 Use Case Diagram

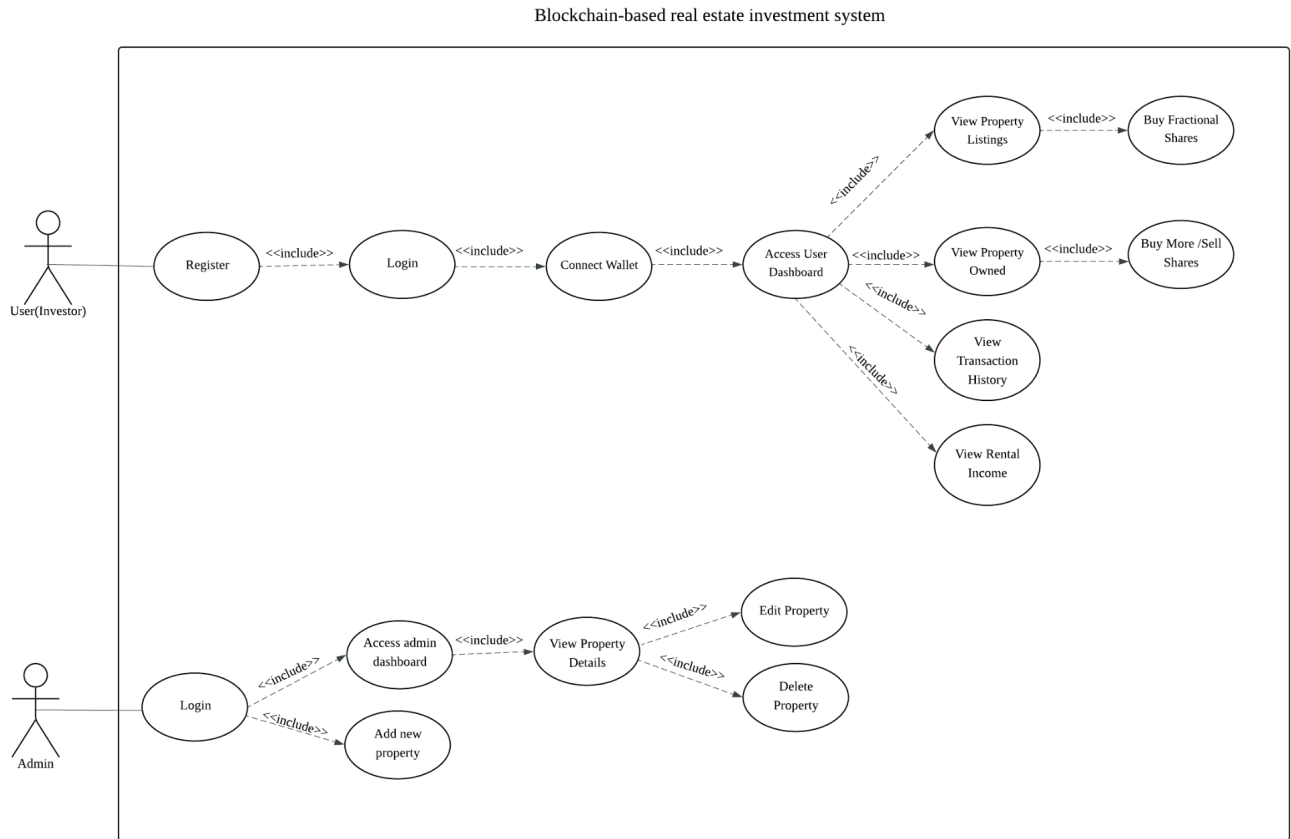


Figure 7.0.1 - Use Case Diagram - Innova

8.0 System Architecture/Design Structure

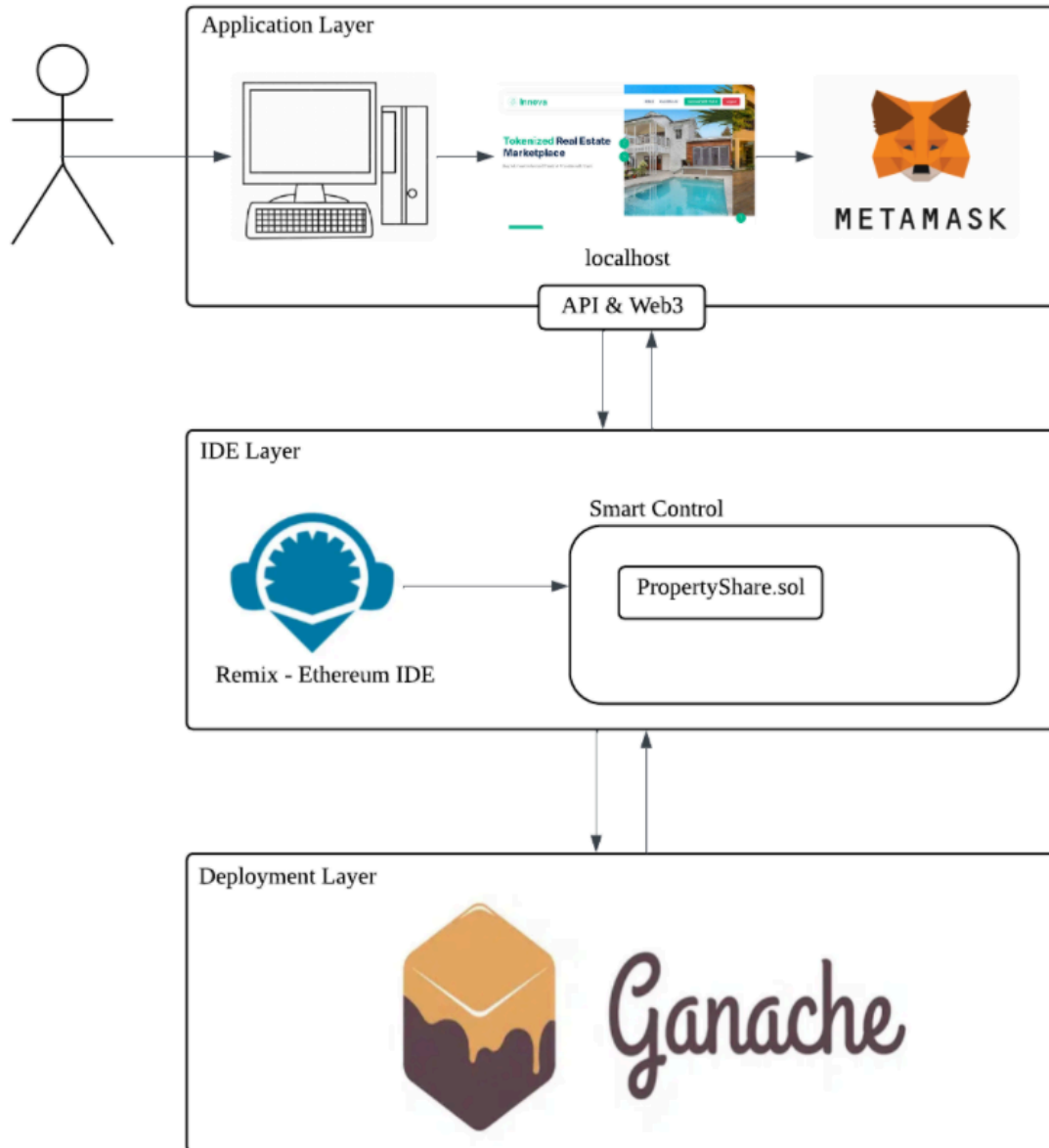


Figure 8.0.1 of System Architecture - Innova

9.0 UI Design

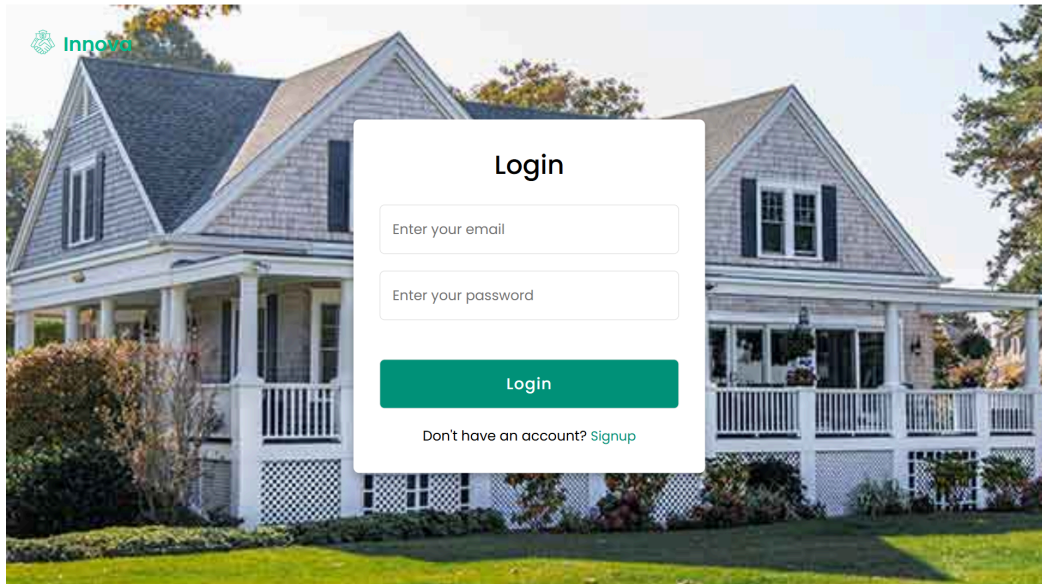


Figure 9.0.1 Login Page

Admin and user need to log in to their account before accessing any page.

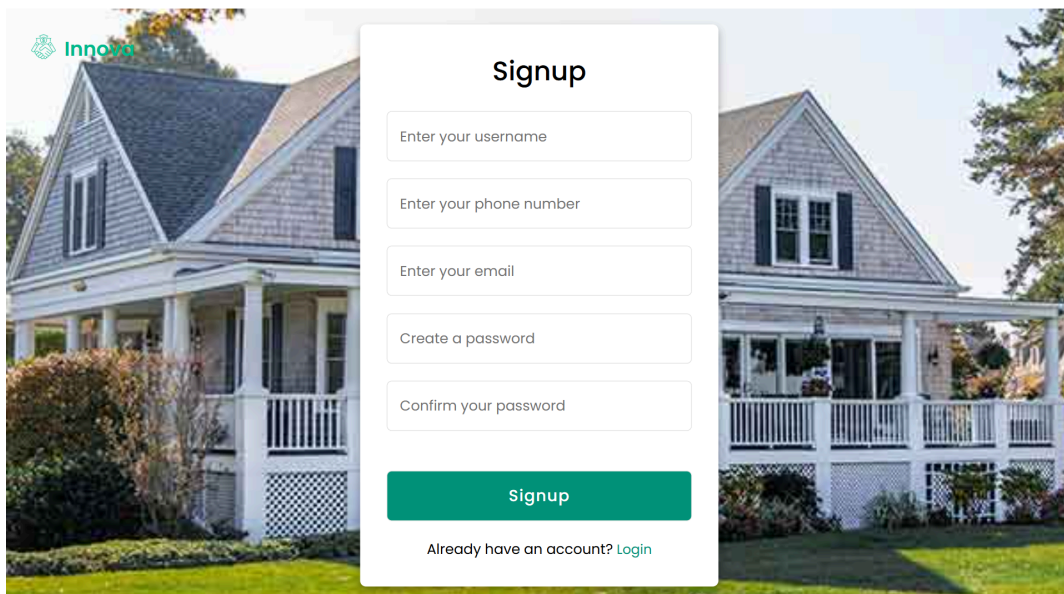


Figure 9.0.2 Sign Up Page

Users can sign up for their account on this page.

9.0.1 Admin

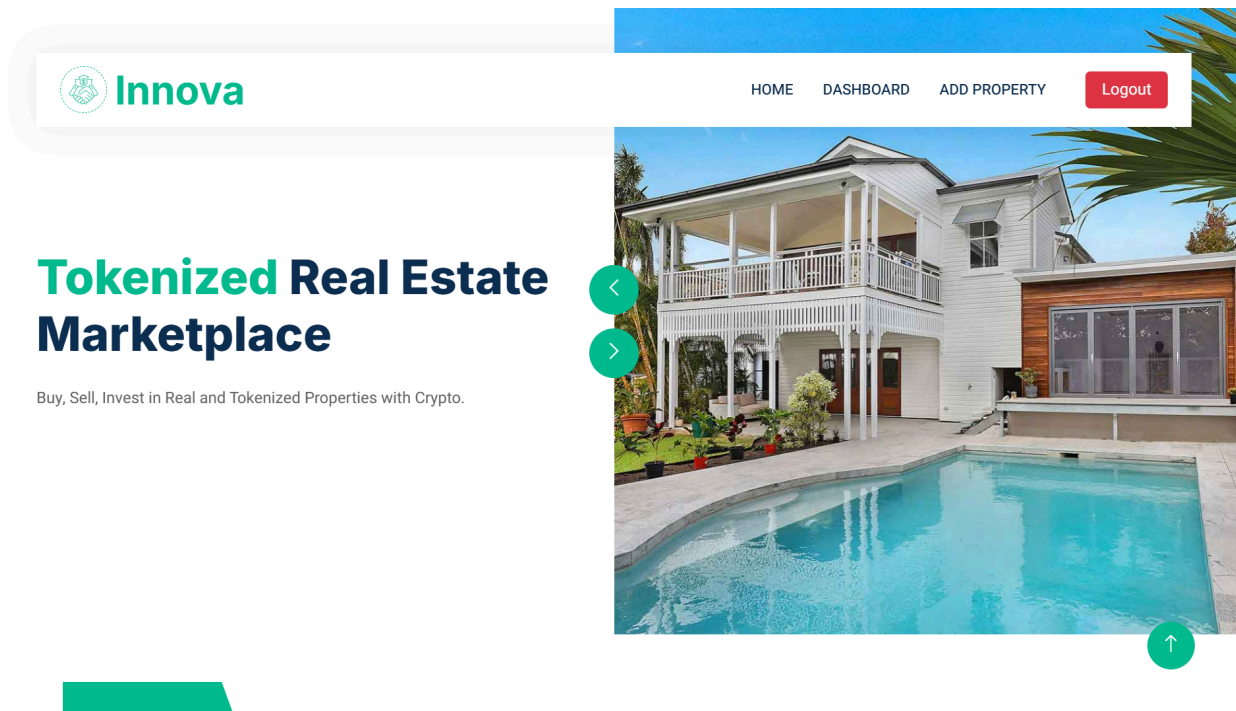


Figure 9.0.1.1 Admin Main Page

After the admin log in their account, it will navigate to this page.

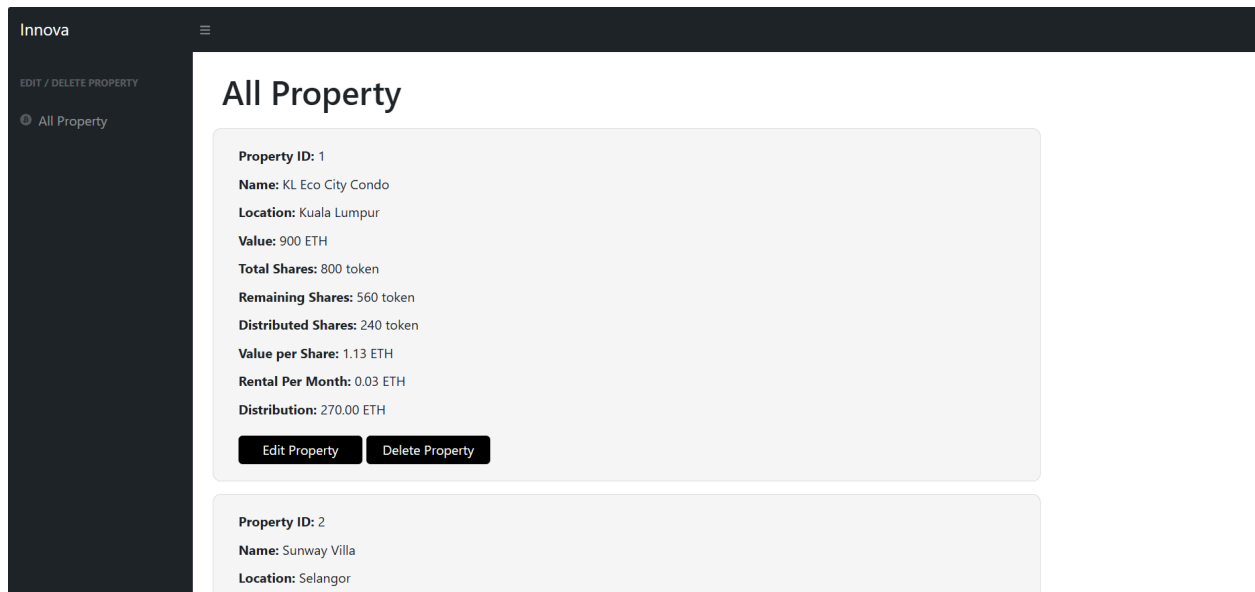


Figure 9.0.1.2 Admin Dashboard Page

Admin can click on the “Dashboard” button on the navigation bar in the admin main page to navigate to this admin dashboard page to view, edit or delete the property. Admin can choose on “Edit Property” or “Delete Property” button to edit or delete the property.

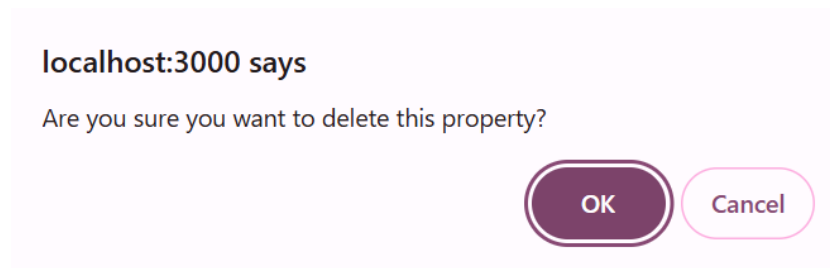
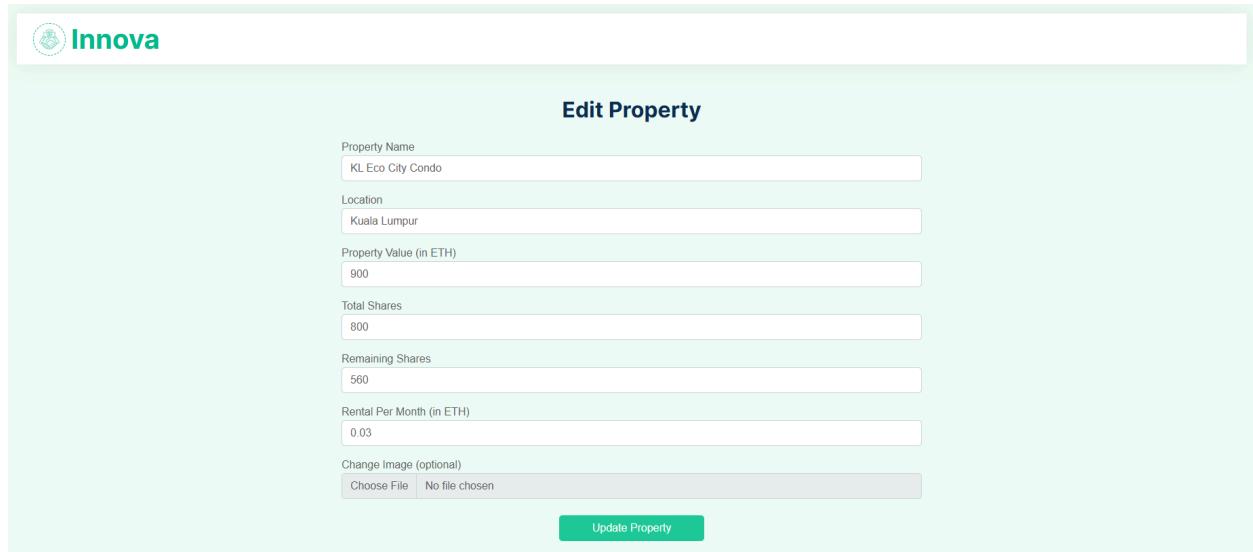


Figure 9.0.1.3 Confirmation Message - Delete Property

After the admin clicks on “Delete Property” button, the system will prompt a confirmation message to confirm that whether the admin sure to delete the property.



Innova

Edit Property

Property Name
KL Eco City Condo

Location
Kuala Lumpur

Property Value (in ETH)
900

Total Shares
800

Remaining Shares
560

Rental Per Month (in ETH)
0.03

Change Image (optional)
Choose File | No file chosen

Update Property

Figure 9.0.1.4 Edit Property Page

After the admin clicks on the “Edit Property” button, it will navigate to this edit property page, and the admin can edit property and click on “Update Property” button after they finish editing the property.

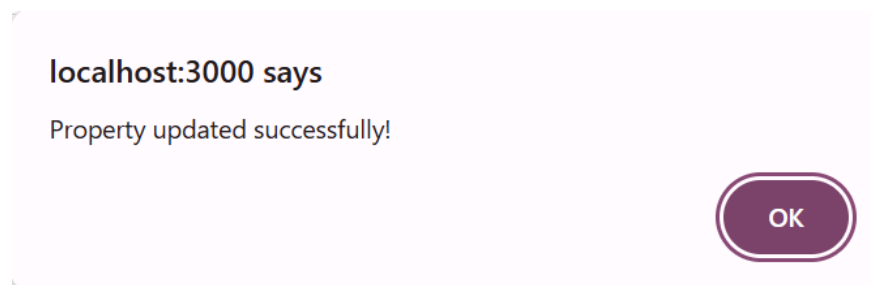
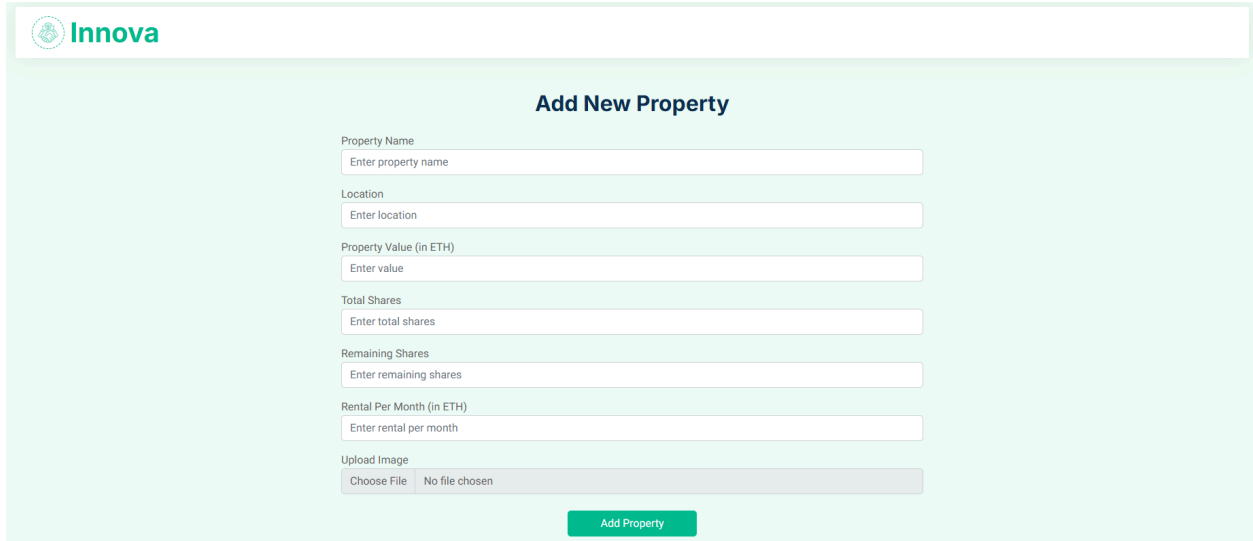


Figure 9.0.1.5 Property Updated Successfully Message

After the admin successfully edit property, it will show this successful message.



The screenshot shows a web interface for adding a new property. At the top left is the 'Innova' logo. The main heading is 'Add New Property'. Below this, there are several input fields: 'Property Name' (with placeholder 'Enter property name'), 'Location' (with placeholder 'Enter location'), 'Property Value (in ETH)' (with placeholder 'Enter value'), 'Total Shares' (with placeholder 'Enter total shares'), 'Remaining Shares' (with placeholder 'Enter remaining shares'), and 'Rental Per Month (in ETH)' (with placeholder 'Enter rental per month'). At the bottom of the form is an 'Upload Image' section with a 'Choose File' button and the text 'No file chosen'. A green 'Add Property' button is located at the bottom right of the form area.

Figure 9.0.1.6 Add Property Page

Admin can click on the “Add Property” button on the navigation bar in the admin main page to access this page. After filling in all the property details, the admin can click on the “Add Property” button to add new property.

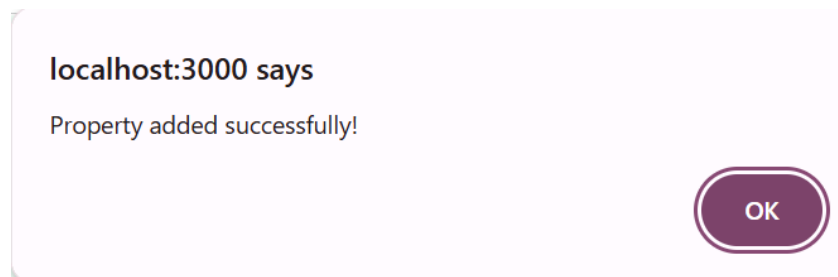


Figure 9.0.1.7 Property Added Successfully Message

After the property is added successfully, it will prompt this successful message.

9.0.2 Investor (User)

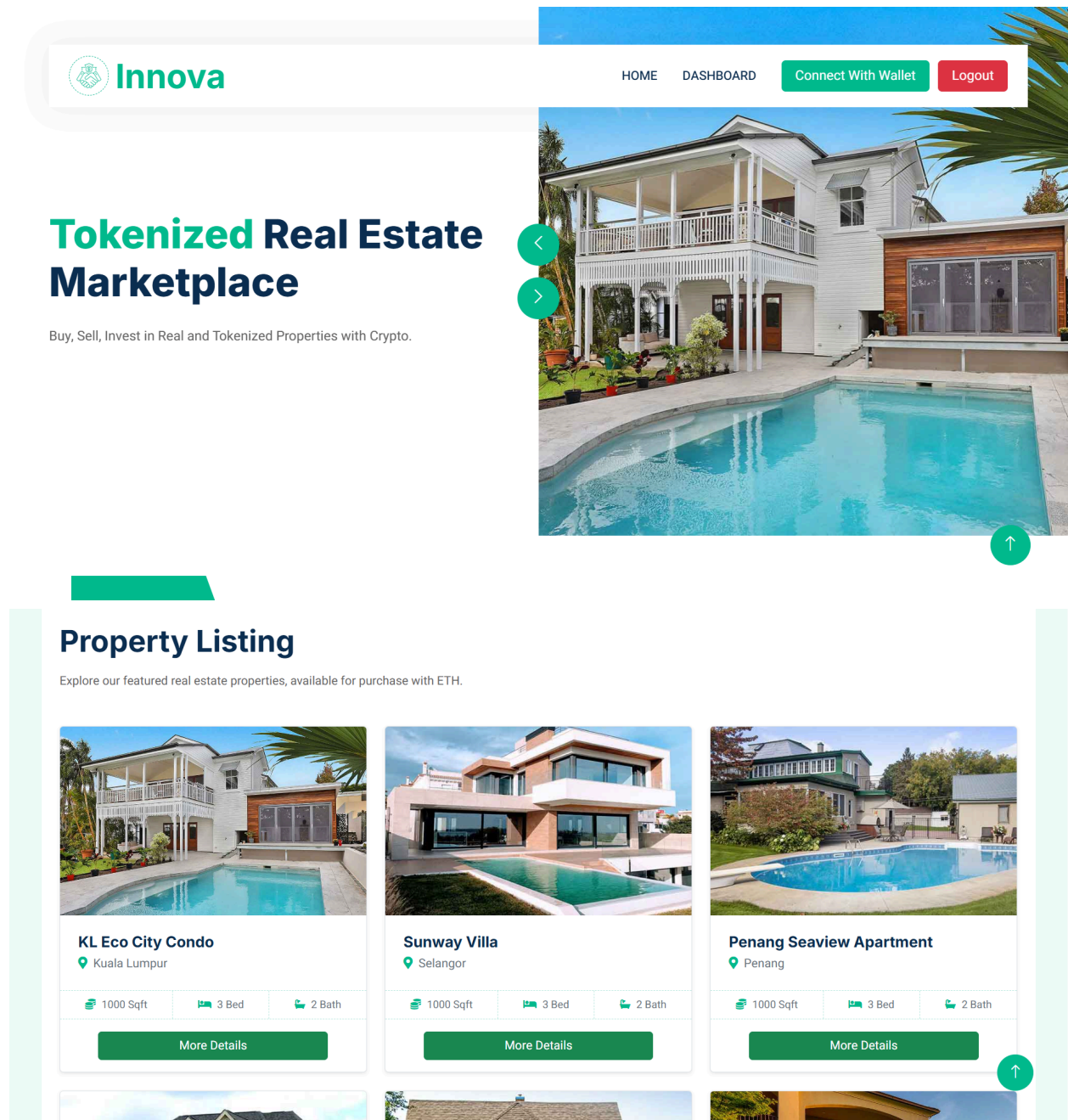


Figure 9.0.2.1, 9.0.2.2 User Main Page

After the user login successfully, it will navigate to this user's main page. Users are required to click on the “Connect With Wallet” button on the navigation bar to connect their wallet before they access to the user dashboard page.

[HOME](#)[DASHBOARD](#)[Disconnect Wallet](#)[Logout](#)

Figure 9.0.2.3 Connect Wallet

After the user connects their wallet, the user can see that the “Connect With Wallet” button will become the “Disconnect Wallet” button. Users can click on it to disconnect their wallet also. Users also can click on the “Dashboard” button to access the dashboard page after they connect their wallet.

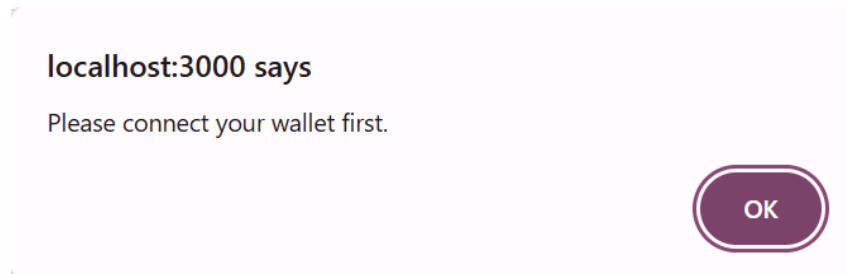


Figure 9.0.2.4 Error Message - User Did Not Connect Wallet

If the user did not connect their wallet before accessing the dashboard page, it will show this error message.

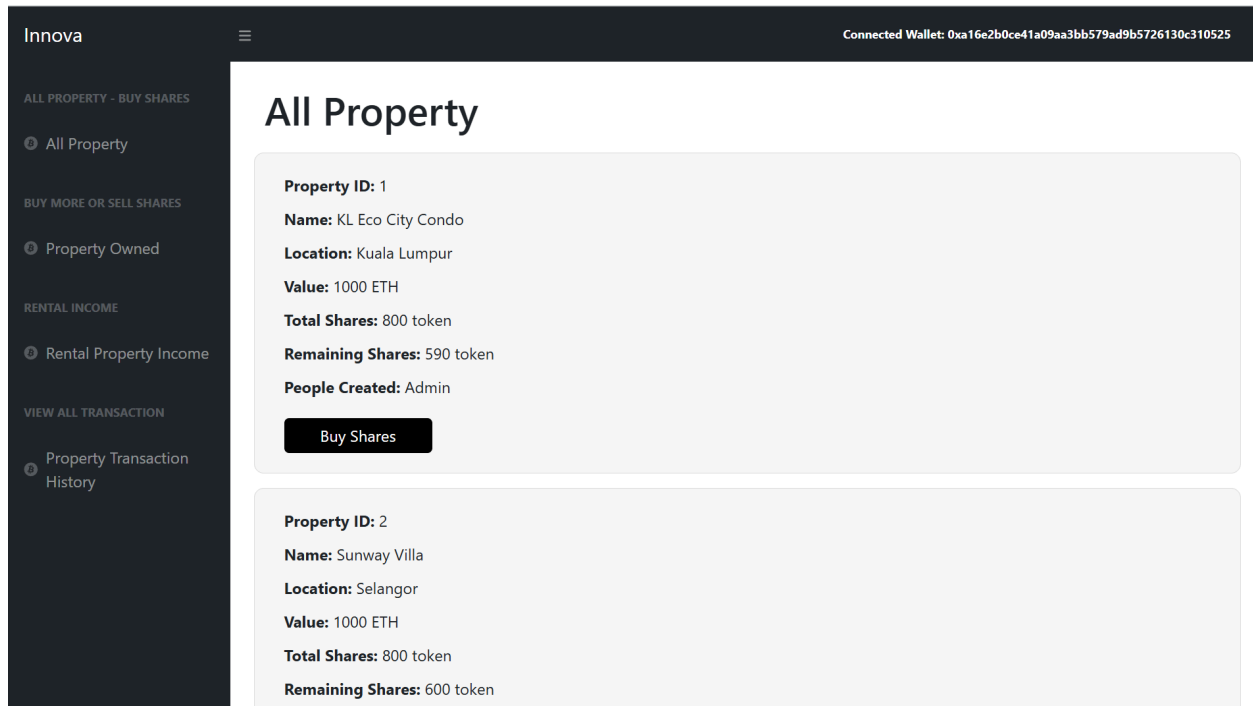


Figure 9.0.2.5 User Dashboard Page

This is the user dashboard page where users can buy fractional shares, buy more or sell their shares and view their transaction history. It will list all the properties that are available to let the user buy the shares, users just need to click on the “Buy Shares” button to buy.

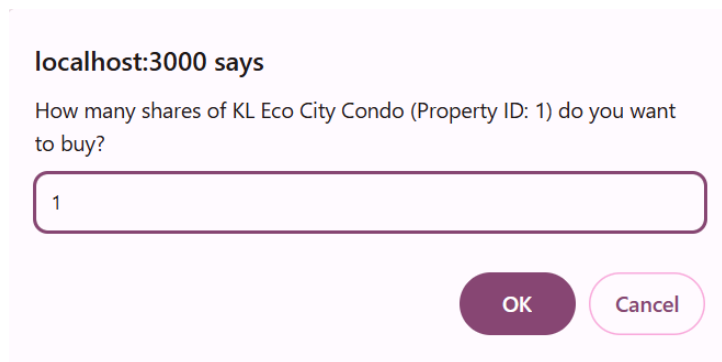


Figure 9.0.2.6 Enter Amount To Buy Shares

After clicking on the “Buy Shares” button, it will ask users to enter the amount of shares that they want to buy.

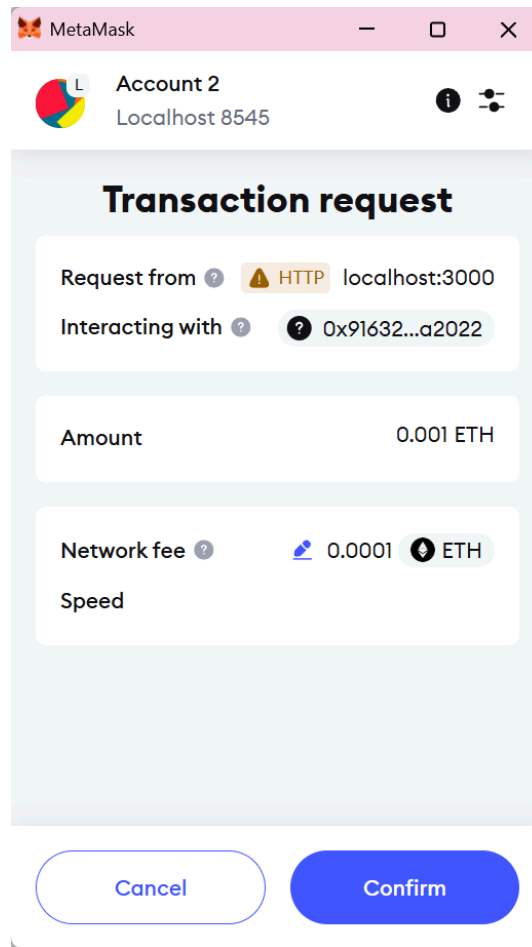


Figure 9.0.2.7 MetaMask Confirmation For Buy Share

After the user enters the amount of shares, the metamask will prompt confirmation for buying the shares. Users just need to click on the “Confirm” button to confirm the buying process.

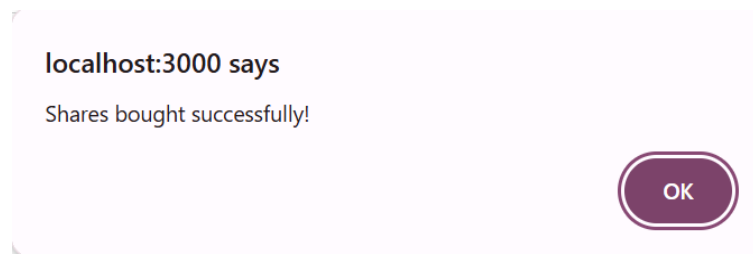


Figure 9.0.2.8 Shares Bought Successfully Message

After successfully buying shares, it will prompt the successful message.

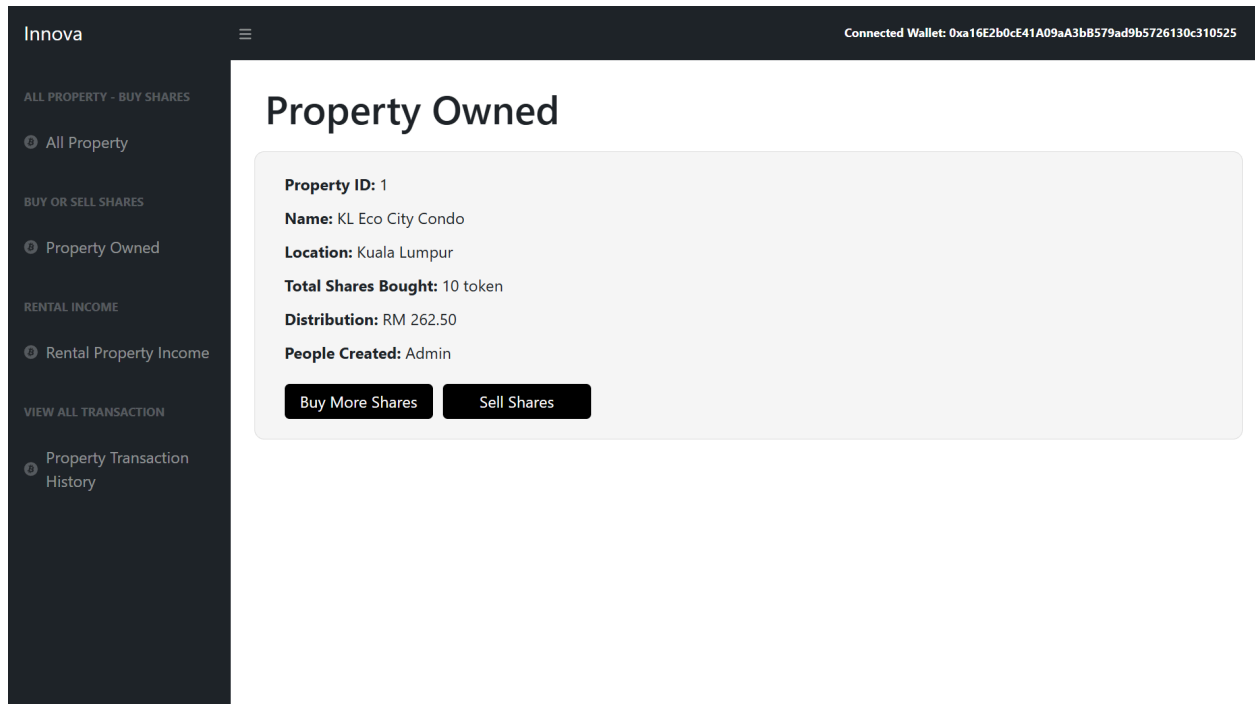


Figure 9.0.2.9 PropertyOwned Page

In this page, it will show all the property that the user bought shares. Users can click on the “Buy More Shares” button to buy more shares. The process of buying more shares will be the same as the buy shares process. Users also can click on the “Sell Shares” button to sell their shares.

The image shows a modal dialog box with a light pink background. The title bar says 'localhost:3000 says'. Below the title, the text reads 'Enter number of shares to sell for KL Eco City Condo (Property ID: 1):'. There is a text input field with a pink border containing the number '1'. At the bottom right of the dialog are two buttons: a dark pink 'OK' button and a light pink 'Cancel' button.

Figure 9.0.2.10 Enter Amount To Sell Shares

After clicking on the “Sell Shares” button, it will ask users to enter the amount of shares that they want to sell.

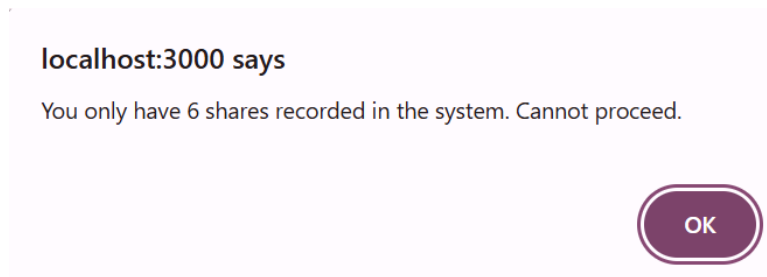


Figure 9.0.2.11 Error Message Of Sell Shares

If the user chooses the amount that is more than the total shares that they bought, it will show an error message.

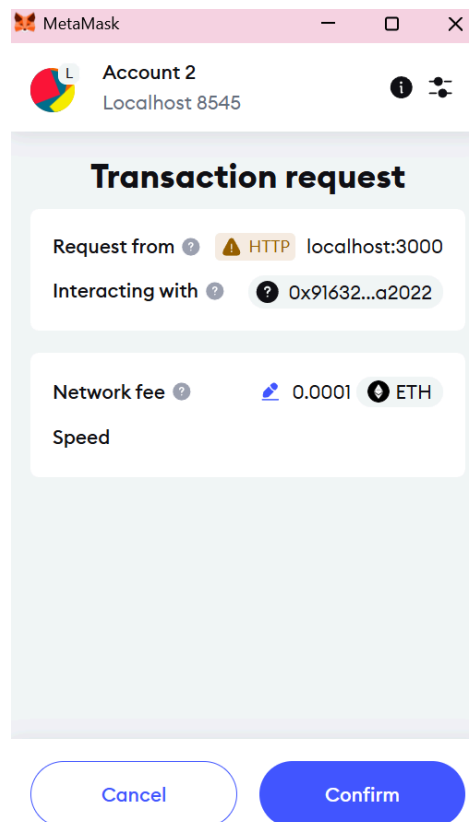


Figure 9.0.2.12 MetaMask Confirmation For Sell Shares

After the user enters the amount of shares that they want to sell, the metamask will prompt confirmation for selling the shares. Users just need to click on the “Confirm” button to confirm the selling process.

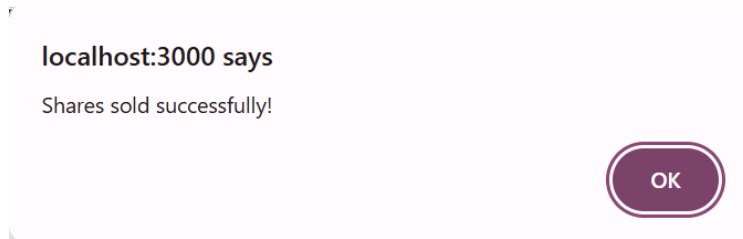


Figure 9.0.2.13 Shares Sold Successfully Message

After successfully selling shares, it will prompt the successful message.

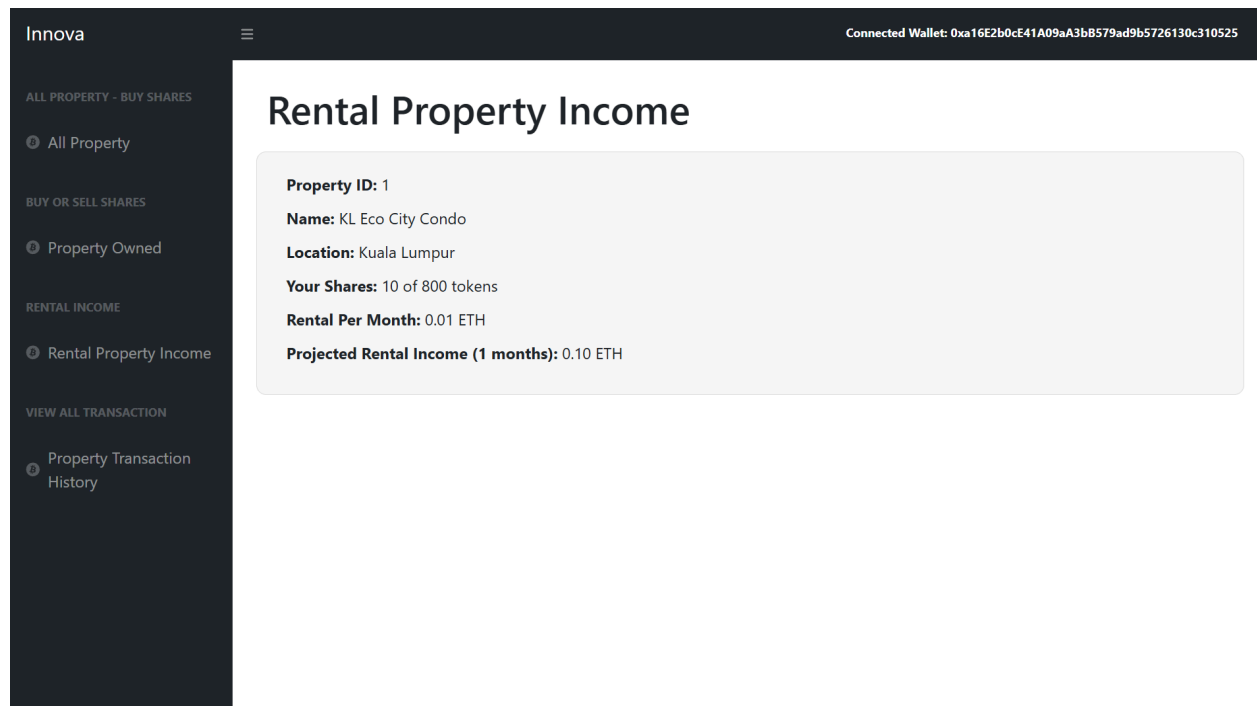


Figure 9.0.2.14 Rental Property Income Page

In this page, it provides users with a detailed overview of the rental earnings generated from their investment in tokenized real estate properties.

Innova

Connected Wallet: 0xa16E2b0cE41A09aA3bB579ad9b5726130c310525

ALL PROPERTY - BUY SHARES

① All Property

BUY OR SELL SHARES

① Property Owned

RENTAL INCOME

① Rental Property Income

VIEW ALL TRANSACTION

① Property Transaction History

Transaction History

Property: KL Eco City Condo

Total Shares: 1 token

Transaction Type (Buy/Sell): Sell

Transaction Time: 2025-05-06 15:38:12

Property: KL Eco City Condo

Total Shares: 1 token

Transaction Type (Buy/Sell): Buy

Transaction Time: 2025-05-06 15:37:55

Property: KL Eco City Condo

Total Shares: 10 token

Transaction Type (Buy/Sell): Buy

Transaction Time: 2025-05-06 15:37:36

Figure 9.0.2.15 Transaction History Page

Users can view all the transactions that they made on this page.

ACCOUNTS

BLOCKS

TRANSACTIONS

CONTRACTS

EVENTS

LOGS

SEARCH FOR BLOCK NUMBERS OR TX HASHES

CURRENT BLOCK 227	GAS PRICE 20000000000	GAS LIMIT 6721975	HARDFORK MERGE	NETWORK ID 5777	RPC SERVER HTTP://127.0.0.1:7545	MINING STATUS AUTOMINING	WORKSPACE REAL ESTATE	SWITCH	
BLOCK 227	MINED ON 2025-05-06 15:38:12					GAS USED 36292	1 TRANSACTION		
BLOCK 226	MINED ON 2025-05-06 15:37:55					GAS USED 29526	1 TRANSACTION		
BLOCK 225	MINED ON 2025-05-06 15:37:36					GAS USED 29526	1 TRANSACTION		
BLOCK 224	MINED ON 2025-05-06 15:34:58					GAS USED 954539	1 TRANSACTION		
BLOCK 223	MINED ON 2025-05-05 17:06:21					GAS USED 36292	1 TRANSACTION		
BLOCK 222	MINED ON 2025-05-05 17:06:10					GAS USED 29526	1 TRANSACTION		
BLOCK 221	MINED ON 2025-05-05 17:03:11					GAS USED 36292	1 TRANSACTION		
BLOCK 220	MINED ON 2025-05-05 17:01:37					GAS USED 36292	1 TRANSACTION		
BLOCK 219	MINED ON 2025-05-05 17:01:14					GAS USED 29526	1 TRANSACTION		
BLOCK 218	MINED ON 2025-05-05 16:57:10					GAS USED 36292	1 TRANSACTION		

Figure 9.0.2.16 Ganache - Blocks

The ganache is used to simulate and record each transaction or smart contract interaction during development, allowing developers to test and debug their blockchain applications in a local environment.

10.0 Tools and Technologies Used

Tools and Technologies	Explanation
Ethereum 	Ethereum is a popular blockchain platform that allows you to design and execute decentralised applications (dApps) using smart contracts (Ethereum, 2025).
Remix IDE (latest version v0.62.0) 	Remix IDE is an online tool for writing, testing, and deploying smart contracts written in Solidity (Remix IDE, 2025).
MetaMask Wallet (latest version 12.15.1) 	MetaMask Wallet is a browser extension that allows you to safely store and manage bitcoin, as well as connect to Ethereum apps (MetaMask, 2025).

Solidity (latest version 0.8.29)



Solidity is a programming language used to create smart contracts on the Ethereum network (Solidity, 2025).

Ganache (latest version v2.7.1)



Ganache is a tool that generates a local Ethereum blockchain on the computer to test your smart contracts (Ganache, 2025).

Truffle (latest release is v24.0)



Truffle is a programming framework that makes it easier to write, test, and deploy Ethereum smart contracts (Ganache, 2025).

11.0 Limitations/challenges faced

Building a blockchain-based platform for real estate fractional ownership was thrilling, but it presented numerous obstacles. Understanding blockchain technology in general was one of the most difficult tasks. Many of our team members were unfamiliar with concepts such as smart contracts, wallets, and how everything links. This made it difficult to construct the system rapidly and correctly, especially because developing smart contracts required specialised knowledge to avoid errors that could lead to security risks.

We also struggled to ensure that our system adhered to the legal standards governing property ownership and data privacy. Because we are dealing with real money and personal information, it is critical to respect the law; nevertheless, blockchain solutions do not automatically satisfy legal requirements such as identity checks. This made our work more complicated.

Another problem concerned the cost and speed of blockchain transactions. Petrol fees on networks such as Ethereum can be extremely costly, particularly when the network is crowded. This makes using the system more expensive for users. Furthermore, public blockchains can only process a finite number of transactions per second, thus during peak times, users may encounter delays, affecting their trust in the network.

Because blockchain keeps data indefinitely, any errors in property details or transactions cannot be quickly corrected. This led us to use extraordinary caution when inputting data. Blockchain apps frequently utilize jargon that non-technical users do not understand, so we had to carefully consider how to construct the system so that it is accessible to everyone. Convincing users to trust and use blockchain for real estate requires additional effort, as many people are still unfamiliar with the technology.

Finally, linking everything; the front-end design, blockchain smart contracts, and back-end database was difficult. Keeping all pieces in sync and ensuring that everything runs well required a significant amount of time and testing.

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